

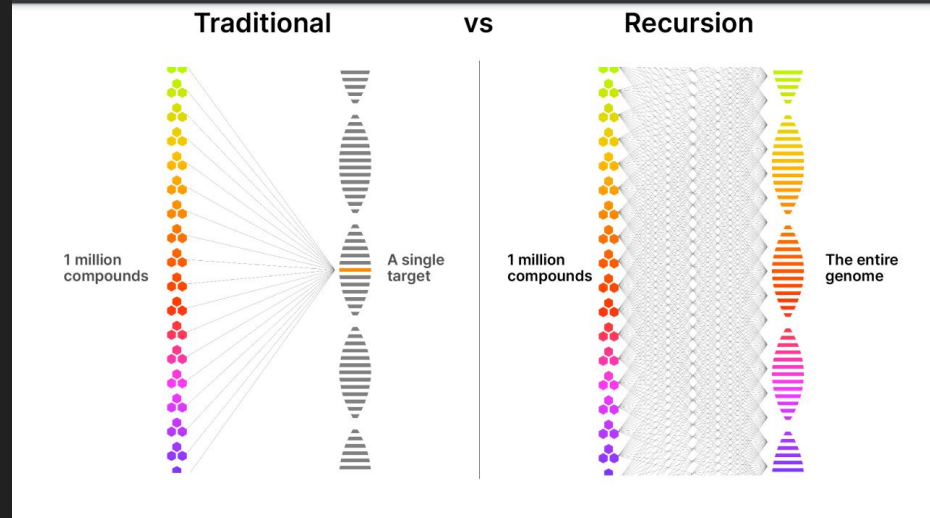
Recursion's Clinical Trial Clearance for solid tumor and lymphoma drug treatment

Biotechnology in the News: October 17th
Rhea Madhogarhia

The News

FDA has given clearance for the clinical trial of a new drug aimed at treating biomarker-enriched solid tumors and lymphoma (blood cancer)

Novelty: Recursion used an AI aided platform for drug discovery, contributing to an expedited research -> clinical trial timeline



The Drug Discovery Process

CDK12 (protein kinase)

- Frequently mutated in ovarian and breast cancer
- Hard to drug because CDK12 inhibitors also tend to inhibit CDK13 resulting in “significant toxicity”

Novel Disease Target: **RBM39 (RNA Binding protein 39)**

- Acts as a pre-mRNA splicing factor
- Promotes exon inclusion
- Overexpressed in many cancers

Identified REC-1245, targets RBM39 and effects CDK12 without affecting CDK13

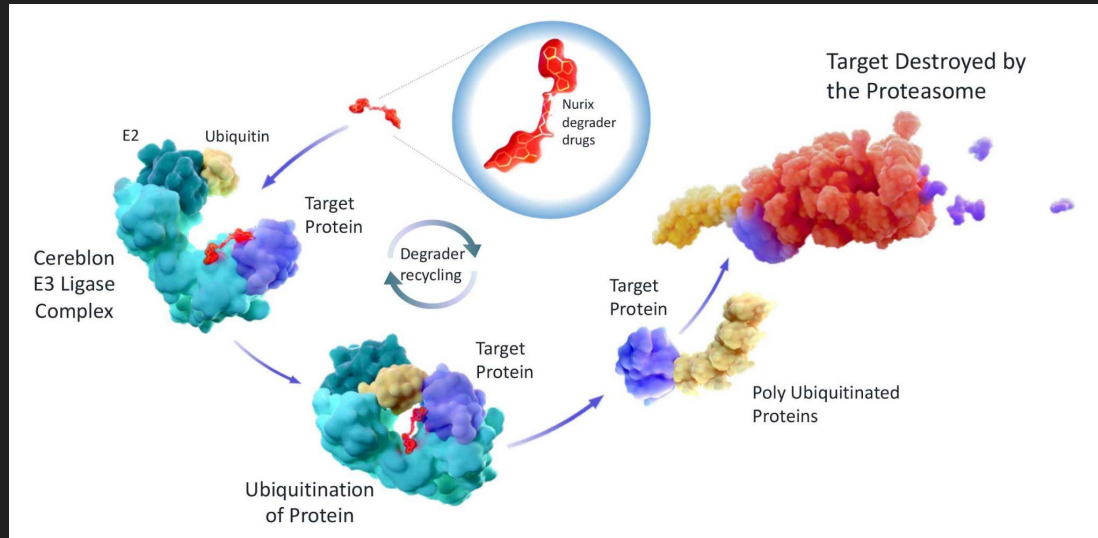


RBM39 and REC-1245

REC-1245 is a RBM39 Degradator (a targeted protein degrader)

RBM39 degradation induces splicing defects which downregulate DNA Damage Response (DDR) networks and cell cycle checkpoints.

Tumor regression



Implications for the biotech field

- New treatment opportunity for individuals with tumors and lymphoma who have been unresponsive to traditional therapies
- More specific targeting of tumors with predictive modeling
- Test new drugs faster!



Sources

<https://www.youtube.com/watch?v=92tLRpKYaWo>

<https://www.biospace.com/press-releases/recursion-announces-fda-clearance-of-investigational-new-drug-application-for-rec-1245-a-potential-first-in-class-rbm39-degrader-for-biomarker-enriched-solid-tumors-and-lymphoma>